POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6034D | Semester VI | Theory | 4 Credits

Electric Vehicles and Traction

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6034D as Electric Vehicles and Traction in Semester VI for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Electric Vehicles course and polytechnic electric traction curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Traction designs, OHE layouts, locomotive circuits and vehicle powertrains must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support traction mechanics, speed-time curves, motor control, and OHE components. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electric Vehicles and Traction studies the systems, mechanics and technology of electric-powered transportation, with a focus on railways and automotive vehicles. It develops the ability to analyze track electrification systems, evaluate traction motor performance using speed-time curves, understand EV powertrain architectures, and coordinate traction power supply solutions while respecting the technical and safety boundaries of electrical and transportation engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Electrical machines, power systems, control systems and basic mechanics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the various systems of track electrification and the mechanics of train movement.
2. Analyze speed-time curves and calculate tractive effort and energy consumption for different services.
3. Evaluate the performance and control of DC and AC motors used in electric traction and vehicles.
4. Explain the components of traction power supply, overhead equipment (OHE) and current collection systems.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the track electrification systems and mechanics of electric traction.	Understand
CO2	Analyze speed-time curves and perform tractive effort calculations.	Apply
CO3	Analyze the characteristics and control of motors used in traction and EVs.	Apply
CO4	Explain the traction power supply, OHE and EV charging infrastructure.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Traction Systems and Mechanics	Systems of traction: Steam, Diesel, Electric; Track electrification systems: DC, Single-phase AC, Three-phase AC, Composite; Mechanics of train movement: Tractive effort, Adhesion, Train resistance.	Approx. 14 hours	CO1	Module 1
2	Speed-Time Curves and Energy Consumption	Speed-time curves for Main line, Suburban and Urban services; Trapezoidal and Quadrilateral approximations; Maximum speed, Average speed, Schedule speed; Specific energy consumption.	Approx. 16 hours	CO2	Module 2
3	Traction Motors and Control	Ideal traction motor requirements; DC series motor characteristics; AC traction motors: Induction and Synchronous; Speed control: Series-parallel control, V/f control; Electrical braking.	Approx. 16 hours	CO3	Module 3
4	Traction Supply and EV Fundamentals	Traction sub-stations; Feeding and distributing system; Overhead Equipment (OHE): Catenary, Contact wire; Current collection: Pantograph, Third rail; EV powertrain and charging basics.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Traction Systems and Mechanics

Systems of traction: Steam, Diesel, Electric; Track electrification systems: DC, Single-phase AC, Three-phase AC, Composite; Mechanics of train movement: Tractive effort, Adhesion, Train resistance.

CO1 · Approx. 14 hours

Speed-Time Curves and Energy Consumption

Speed-time curves for Main line, Suburban and Urban services; Trapezoidal and Quadrilateral approximations; Maximum speed, Average speed, Schedule speed; Specific energy consumption.

CO2 · Approx. 16 hours

Traction Motors and Control

Ideal traction motor requirements; DC series motor characteristics; AC traction motors: Induction and Synchronous; Speed control: Series-parallel control, V/f control; Electrical braking.

CO3 · Approx. 16 hours

Traction Supply and EV Fundamentals

Traction sub-stations; Feeding and distributing system; Overhead Equipment (OHE): Catenary, Contact wire; Current collection: Pantograph, Third rail; EV powertrain and charging basics.

CO4 · Approx. 14 hours

MODULE 1

Traction Systems and Mechanics

Systems of traction: Steam, Diesel, Electric; Track electrification systems: DC, Single-phase AC, Three-phase AC, Composite; Mechanics of train movement: Tractive effort, Adhesion, Train resistance.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

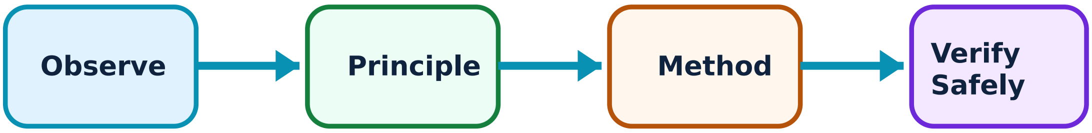
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Traction Systems and Mechanics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Electrification Systems−

Electric traction offers high efficiency and zero emissions at the point of use. Track electrification systems vary by voltage and frequency. DC systems (e.g., 1500V/3000V) are common for metros, while single-phase AC (25kV, 50Hz) is the standard for long-distance railways in India.

Study and application method

Compare 'DC' and 'AC' track electrification systems in a conceptual table; explain the advantages of '25kV AC' traction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'DC' and 'AC' track electrification systems in a conceptual table; explain the advantages of '25kV AC' traction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Traction systems involve extremely high voltages and currents. Do not approach OHE lines or traction equipment without authorised safety clearance and grounding.

2. Traction Mechanics−

Tractive effort is the force required to move the train. It must overcome train resistance (friction, air resistance), gravity (on gradients), and inertia (during acceleration). The 'Coefficient of Adhesion' determines the maximum force that can be applied without wheel slip.

Study and application method

Define 'Tractive Effort'; list the three components of 'Train Resistance'; explain the concept of 'Adhesion' in traction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

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Common mistake / precaution: Wheel slip can damage both the rails and the traction motors. All traction control systems must use authorised anti-slip and anti-skid protection.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Speed-Time Curves and Energy Consumption

Speed-time curves for Main line, Suburban and Urban services; Trapezoidal and Quadrilateral approximations; Maximum speed, Average speed, Schedule speed; Specific energy consumption.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

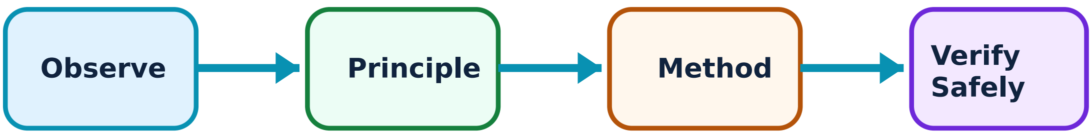
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Speed-Time Curves and Energy Consumption: identify the system, state its principle, follow the correct method and verify the result safely.

1. Speed-Time Curve Analysis−

Speed-time curves represent train movement. Main line services have long free-running periods, while urban/suburban services have frequent starts and stops. Trapezoidal and quadrilateral approximations simplify calculations for tractive effort and energy.

Study and application method

Sketch and compare the 'Speed-Time Curves' for Main line and Suburban services; define 'Schedule Speed' and factors affecting it.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch and compare the 'Speed-Time Curves' for Main line and Suburban services; define 'Schedule Speed' and factors affecting it.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Operational speed limits are strictly regulated for safety. Do not use theoretical speed-time curves for actual train scheduling without authorised safety margin analysis.

2. Energy Consumption−

Specific Energy Consumption (SEC) is the energy consumed per tonne-km of train movement. It depends on the distance between stops, acceleration/braking rates, gradient, and train resistance. Regenerative braking significantly reduces SEC in urban services.

Study and application method

Define 'Specific Energy Consumption'; explain how 'Regenerative Braking' improves the efficiency of a traction system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Specific Energy Consumption'; explain how 'Regenerative Braking' improves the efficiency of a traction system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Energy consumption calculations are based on idealized models. Actual energy use depends on driver behavior and environmental conditions; always follow authorised energy-saving protocols.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Traction Motors and Control

Ideal traction motor requirements; DC series motor characteristics; AC traction motors: Induction and Synchronous; Speed control: Series-parallel control, V/f control; Electrical braking.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

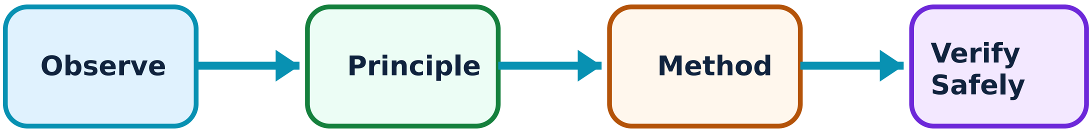
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Traction Motors and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Motors for Traction and EVs−

Traction motors require high starting torque and a wide speed range. DC series motors were traditionally used for their excellent torque characteristics. Modern locomotives and EVs use 3-phase Induction motors or PMSMs due to their robustness and high power density.

Study and application method

List the requirements of an 'Ideal Traction Motor'; compare 'DC Series' and 'AC Induction' motors for traction applications.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the requirements of an 'Ideal Traction Motor'; compare 'DC Series' and 'AC Induction' motors for traction applications.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

> **Malayalam support:** □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

> **Common mistake / precaution:** Traction motors operate under heavy thermal and mechanical stress. Do not exceed authorised motor current and temperature limits to prevent insulation failure.

2. Speed Control and Braking−

Series-parallel control (for DC) and V/f control (for AC) are used to regulate speed efficiently. Electrical braking methods like 'Plugging', 'Rheostatic', and 'Regenerative' braking provide smooth deceleration and reduce mechanical wear on brake shoes.

Study and application method

Explain the 'Series-Parallel' control of two DC traction motors; describe the principle of 'Regenerative Braking' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'Series-Parallel' control of two DC traction motors; describe the principle of 'Regenerative Braking' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Braking failure in traction is a major safety risk. All electrical braking systems must be supplemented by authorised fail-safe mechanical air/vacuum brakes.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Traction Supply and EV Fundamentals

Traction sub-stations; Feeding and distributing system; Overhead Equipment (OHE): Catenary, Contact wire; Current collection: Pantograph, Third rail; EV powertrain and charging basics.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

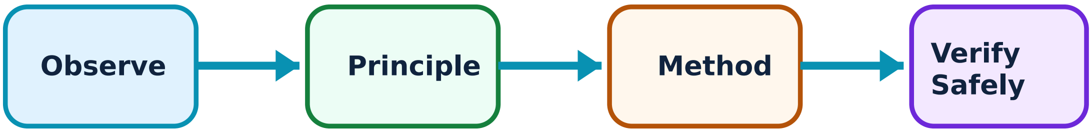
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Traction Supply and EV Fundamentals: identify the system, state its principle, follow the correct method and verify the result safely.

1. Supply and Current Collection−

Power is supplied to locomotives via traction sub-stations. Overhead Equipment (OHE) consists of a catenary wire and a contact wire. Current is collected using a Pantograph. Metros often use a 'Third Rail' system for current collection.

Study and application method

Sketch the components of 'Overhead Equipment' (OHE); explain the function of a 'Pantograph' in an electric locomotive.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the components of 'Overhead Equipment' (OHE); explain the function of a 'Pantograph' in an electric locomotive.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: OHE systems are prone to lightning strikes and mechanical damage. All traction supply systems must follow authorised surge protection and periodic maintenance schedules.

2. Electric Vehicle (EV) Technology−

EVs use on-board batteries to power electric motors. The powertrain includes the battery pack, inverter/controller, and motor. Charging infrastructure includes AC (Level 1/2) and DC Fast chargers (Level 3). BMS ensures battery safety and longevity.

Study and application method

Differentiate between 'BEV' and 'HEV' powertrain architectures; list three common 'EV Battery' chemistries.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'BEV' and 'HEV' powertrain architectures; list three common 'EV Battery' chemistries.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: EV batteries pose high-voltage and thermal runaway risks. Do not perform maintenance on EV battery packs without authorised high-voltage safety training and PPE.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Traction Systems and Mechanics

Use the concept flow to revise observation, principle, method and verification.

Module 2: Speed-Time Curves and Energy Consumption

Use the concept flow to revise observation, principle, method and verification.

Module 3: Traction Motors and Control

Use the concept flow to revise observation, principle, method and verification.

Module 4: Traction Supply and EV Fundamentals

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Trapezoidal and Quadrilateral speed-time curves in a conceptual table.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the current collection system (Pantograph) of an electric locomotive.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of an Electric Vehicle powertrain showing the battery, controller, and motor.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the tractive effort conceptually for a train on a given gradient.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the type of electrification (OHE or Third rail) used in a local railway or metro system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Traction Systems and Mechanics

Explain Electrification Systems and state one correct method, application or precaution.−

Electric traction offers high efficiency and zero emissions at the point of use. Track electrification systems vary by voltage and frequency. DC systems (e.g., 1500V/3000V) are common for metros, while single-phase AC (25kV, 50Hz) is the standard for long-distance railways in India.

Answer framework: Compare 'DC' and 'AC' track electrification systems in a conceptual table; explain the advantages of '25kV AC' traction.

Important point: Traction systems involve extremely high voltages and currents. Do not approach OHE lines or traction equipment without authorised safety clearance and grounding.

Explain Traction Mechanics and state one correct method, application or precaution.−

Tractive effort is the force required to move the train. It must overcome train resistance (friction, air resistance), gravity (on gradients), and inertia (during acceleration). The 'Coefficient of Adhesion' determines the maximum force that can be applied without wheel slip.

Answer framework: Define 'Tractive Effort'; list the three components of 'Train Resistance'; explain the concept of 'Adhesion' in traction.

Important point: Wheel slip can damage both the rails and the traction motors. All traction control systems must use authorised anti-slip and anti-skid protection.

Module 2 — Speed-Time Curves and Energy Consumption

Explain Speed-Time Curve Analysis and state one correct method, application or precaution.−

Speed-time curves represent train movement. Main line services have long free-running periods, while urban/suburban services have frequent starts and stops. Trapezoidal and quadrilateral approximations simplify calculations for tractive effort and energy.

Answer framework: Sketch and compare the 'Speed-Time Curves' for Main line and Suburban services; define 'Schedule Speed' and factors affecting it.

Important point: Operational speed limits are strictly regulated for safety. Do not use theoretical speed-time curves for actual train scheduling without authorised safety margin analysis.

Explain Energy Consumption and state one correct method, application or precaution.

−

Specific Energy Consumption (SEC) is the energy consumed per tonne-km of train movement. It depends on the distance between stops, acceleration/braking rates, gradient, and train resistance. Regenerative braking significantly reduces SEC in urban services.

Answer framework: Define 'Specific Energy Consumption'; explain how 'Regenerative Braking' improves the efficiency of a traction system.

Important point: Energy consumption calculations are based on idealized models. Actual energy use depends on driver behavior and environmental conditions; always follow authorised energy-saving protocols.

Module 3 — Traction Motors and Control

Explain Motors for Traction and EVs and state one correct method, application or precaution.−

Traction motors require high starting torque and a wide speed range. DC series motors were traditionally used for their excellent torque characteristics. Modern locomotives and EVs use 3-phase Induction motors or PMSMs due to their robustness and high power density.

Answer framework: List the requirements of an 'Ideal Traction Motor'; compare 'DC Series' and 'AC Induction' motors for traction applications.

Important point: Traction motors operate under heavy thermal and mechanical stress. Do not exceed authorised motor current and temperature limits to prevent insulation failure.

Explain Speed Control and Braking and state one correct method, application or precaution.−

Series-parallel control (for DC) and V/f control (for AC) are used to regulate speed efficiently. Electrical braking methods like 'Plugging', 'Rheostatic', and 'Regenerative' braking provide smooth deceleration and reduce mechanical wear on brake shoes.

Answer framework: Explain the 'Series-Parallel' control of two DC traction motors; describe the principle of 'Regenerative Braking' conceptually.

Important point: Braking failure in traction is a major safety risk. All electrical braking systems must be supplemented by authorised fail-safe mechanical air/vacuum brakes.

Module 4 — Traction Supply and EV Fundamentals

Explain Supply and Current Collection and state one correct method, application or

precaution.−

Power is supplied to locomotives via traction sub-stations. Overhead Equipment (OHE) consists of a catenary wire and a contact wire. Current is collected using a Pantograph. Metros often use a 'Third Rail' system for current collection.

Answer framework: Sketch the components of 'Overhead Equipment' (OHE); explain the function of a 'Pantograph' in an electric locomotive.

Important point: OHE systems are prone to lightning strikes and mechanical damage. All traction supply systems must follow authorised surge protection and periodic maintenance schedules.

Explain Electric Vehicle (EV) Technology and state one correct method, application or precaution.−

EVs use on-board batteries to power electric motors. The powertrain includes the battery pack, inverter/controller, and motor. Charging infrastructure includes AC (Level 1/2) and DC Fast chargers (Level 3). BMS ensures battery safety and longevity.

Answer framework: Differentiate between 'BEV' and 'HEV' powertrain architectures; list three common 'EV Battery' chemistries.

Important point: EV batteries pose high-voltage and thermal runaway risks. Do not perform maintenance on EV battery packs without authorised high-voltage safety training and PPE.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Traction Systems and Mechanics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Speed-Time Curves and Energy Consumption.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Traction Motors and Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Traction Supply and EV Fundamentals.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Systems of traction: Steam, Diesel, Electric; Track electrification systems: DC, Single-phase AC, Three-phase AC, Composite; Mechanics of train movement: Tractive effort, Adhesion, Train resistance..
2. Develop a complete answer or practical plan for: Speed-time curves for Main line, Suburban and Urban services; Trapezoidal and Quadrilateral approximations; Maximum speed, Average speed, Schedule speed; Specific energy consumption..
3. Develop a complete answer or practical plan for: Ideal traction motor requirements; DC series motor characteristics; AC traction motors: Induction and Synchronous; Speed control: Series-parallel control, V/f control; Electrical braking..
4. Develop a complete answer or practical plan for: Traction sub-stations; Feeding and distributing system; Overhead Equipment (OHE): Catenary, Contact wire; Current collection: Pantograph, Third rail; EV powertrain and charging basics..

LAST-MILE REVISION

Quick Revision

Module 1: Traction Systems and Mechanics

Systems of traction: Steam, Diesel, Electric; Track electrification systems: DC, Single-phase AC, Three-phase AC, Composite; Mechanics of train movement: Tractive effort, Adhesion, Train resistance.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Speed-Time Curves and Energy Consumption

Speed-time curves for Main line, Suburban and Urban services; Trapezoidal and Quadrilateral approximations; Maximum speed, Average speed, Schedule speed; Specific energy consumption.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Traction Motors and Control

Ideal traction motor requirements; DC series motor characteristics; AC traction motors: Induction and Synchronous; Speed control: Series-parallel control, V/f control; Electrical braking.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Traction Supply and EV Fundamentals

Traction sub-stations; Feeding and distributing system; Overhead Equipment (OHE): Catenary, Contact wire; Current collection: Pantograph, Third rail; EV powertrain and charging basics.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Electrification Systems	Electric traction offers high efficiency and zero emissions at the point of use.
Traction Mechanics	Tractive effort is the force required to move the train.
Speed-Time Curve Analysis	Speed-time curves represent train movement.
Energy Consumption	Specific Energy Consumption (SEC) is the energy consumed per tonne-km of train movement.
Motors for Traction and EVs	Traction motors require high starting torque and a wide speed range.
Speed Control and Braking	Series-parallel control (for DC) and V/f control (for AC) are used to regulate speed efficiently.
Supply and Current Collection	Power is supplied to locomotives via traction sub-stations.
Electric Vehicle (EV) Technology	EVs use on-board batteries to power electric motors.

LEARNING RESOURCES

References

Recommended traction and EV references

1. H. Partab, Modern Electric Traction.
2. J. B. Gupta, Utilization of Electric Power and Electric Traction.
3. G. C. Garg, Utilization of Electric Power and Electric Traction.
4. Ashok Jhunjhunwala, Electric Vehicles and Renewable Energy.

Approved public traction and EV sources

- https://nptel.ac.in/courses/108102121
- https://onlinecourses.nptel.ac.in/noc25_ee33/preview

These sources provide the verified study basis while official Course 6034D detail is unavailable; they do not replace official locomotive designs, authorised railway safety protocols, certified equipment specifications or authorised transport plans.